ANJANA ANAND

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PROFESSIONAL SUMMARY

- 5 years of IT experience in telecommunication, retail, and health care industries.
- 3+ years of experience in Data Science including Machine Learning, Data Mining and Statistical Analysis.
- 1+ years of experience in Containerization (Kubernetes and docker) deploying build applications in on-prem cloud.
- 1+ years of experience in Web Services and REST API to integrate Web-based applications using XML, SOAP and JSON.
- Experience in performing exploratory data analysis, dimensionality reduction and model evaluation.
- Capable in Text Analytics, developing different Statistical Machine Learning, Data mining solutions to various business problems, and generating data visualizations using Python and Grafana.
- Proficient in regression algorithms which includes Linear Regression, Logistic Regression, Poisson Regression, Lasso and Ridge Regression.
- Working knowledge in classification algorithms which includes Decision Tree, K-nearest neighbour(KNN), Naïve Bayes, Support Vector Machines(SVM), Convolutional Neural Network (CNN), ensemble methods like Bagging, Boosting, Random Forest and SMOTE.
- Working knowledge in clustering algorithms like Hierarchical clustering, K means and Density Based clustering.
- Proficient in developing project (SDLC) in Agile and Waterfall methodology. .
- Good knowledge on python libraries which includes NumPy, Pandas, Matplotlib, Seaborn, Keras, TensorFlow, NLTK and Scikit-Learn.
- Experience in performing data analysis on various IDE's like Jupyter Notebook, Spyder and PyCharm.
- Knowledge of SQL Queries and optimizing the queries in SQL Server, and MySQL.
- Good knowledge in technologies like HDFS, Hadoop map reduce, Horton works and AWS EMR, AWS Sagemaker.
- Good Knowledge on Version control systems such as Git and Bitbucket.

WORK EXPERIENCE

Qualcomm, Remote – Minneapolis - IT Engineer in Data Science **Jul 2020 – Present**

Event management solution with Moogsoft to implement grouping of alerts and incidents with Build Services and applications to improve alert detection and remediation.

- Analysis of large volume of IT telemetry data from alerting and monitoring sources like App Dynamic, Sitescope, Splunk, eHealth and Service Now.
- Inbound integration with target systems with Rest API endpoints using python and tool-based integration.

- Performed data processing steps which includes data cleaning, data aggregation and dimensionality reduction.
- Integration with Machine Database to perform service mapping for alert enrichment that facilitates better clustering of alerts using python.
- Deduplication of the data to reduce operational noise with python.
- Performed feature selection using the corelation coefficient between the event attributes.
- Implemented clustering model to calculate the textual similarity between events using vertex entropy and hop limit to cluster alerts into meaningful groups.
- Created multiple ML recipes in Moogsoft and compared each other for better performance.
- Incremental improvement of accuracy is ensured by providing manual validations of the clustering with application/product owners.
- Developed KPIs and created graphical and visual representation with Splunk and Grafana to present complex data into easily understandable format for leadership teams.
- Achieved 94% noise reduction that improved the overall detection and remediation of incidents, ensuring continuous service delivery on business level.
- Good experience in alert generation and alert monitoring in Splunk and interaction with the databases using SQL.
- Communication and coordination with cross-functional and global teams to achieve overall quality and timeliness of the project deliverables.
- Maintained the code and versions with GitHub for version control.
- Experience in Agile environment to deliver projects incrementally and make rapid adjustments as needed.

Application deployment and administration using Kubernetes.

- Leading administration, development, and support of build applications.
- Working excessively in Linux, Docker, and Kubernetes environment to create an infrastructure for build applications like Jenkins and Electric commander.
- Experience in creating/deploying singleton and replica set application using Rancher (Kubernetes management platform) in on-prem cloud with CIFS/NFS volumes as persistent storage and stateful-set as the deployment model.
- Automated the deployment in Jenkin with bash script.
- Experience in creating build analytics through Splunk dashboards.

Target Corporation, Minneapolis - Data Scientist Consultant **Dec 2018 – March 2020**

Implemented text classification solution to process the characteristics of items and determine if the item can be categorized as one of 3 categories: 1. Environmentally sensitive 2. Nonenvironmentally sensitive 3. e-waste. This information is required in stores/supply chain for handling the items for disposal.

- Connection to the PostgreSQL database to fetch the item data using psycopg2 package in python.
- Worked on complex Sql queries to retrieve the required item details from the database.
- Performed Data Cleaning, features scaling, features selection using Pandas and NumPy packages in python.
- Performed extensive text processing of the item description using tokenization, lemmatization, stemming and stop word removal techniques.
- Vectorization of the item description data was performed by tf-idf and word2vec approaches with feature extraction using unigrams and bigrams.
- Exploratory data analysis using seaborn and matplotlib to compute the descriptive statistics.

- Dealt with imbalanced class by using sampling techniques like down-sampling, up-sampling based on the item department type.
- Setup dedicated training environment with suitable compute resources in Linux machines.
- Used Machine learning and deep learning techniques which includes Random Forest, Naïve Bayes and CNN models to automate categorization of the items based on the item characteristics.
- Built confusion matrix and plotted the precision & recall scores of the created model for identifying ideal classification threshold.
- Incremental improvement of accuracy is ensured by using ensemble techniques and providing manual validations of the predictions.
- Automated the triggering of scripts on demand through CI pipelines from GitHub using Jenkins.
- Compared metrics of accuracy for model predictions in Postgres database and create visualization in Grafana.
- Alerts from Grafana when a specific department items consistently have a poor performance & identify cases for re-training.
- Exposed the models as API for consuming live data via web app
- This eliminated the manual effort to review and categorize each individual item. With this implementation, from the item load of ~50K, only items around ~2K is being manually reviewed every week.
- Performed POC for the same project on AWS Sage Maker and presented the results to the team.

Health Partners, Minneapolis - Data Scientist Consultant **Mar 2017 – July 2018**

Automatic classification of high-risk patience for sending periodic notification on health checkups.

- Performed data collection by merging demographic and medical history data of the patients using complex SQL query from the MySQL database.
- Exploratory data analysis through statistical insights using python, seaborn, matplotlib to study the behavior of the patients and medical conditions.
- Implemented Principal component analysis (PCA) for feature extraction to select the most important features.
- Addressed overfitting and underfitting by tuning the hyper parameter of the algorithm and by using L1 and L2 Regularization.
- Implemented cross validation method for data partition with the ratio of 80%-20% partitioning.
- Worked on classification algorithms like Logistic regression, K-Nearest Neighbor, Decision tree and Support vector.
- Handled class imbalance problem by creating synthetic samples of data using Synthetic Minority Over-Sampling Technique (SMOTE).
- Computed confusion matrix and compared the recall, accuracy and F1 score to find the best performing model.
- This classification is then used by the member service team to send periodic notification and reminders to patients at high risk.

Big data processing of the paid media Google search ads to compute various statistics on the user activity. These results were consumed by the marketing team for making critical business decisions.

- Hadoop MapReduce implementation for parallel processing of paid media data and produce statistics on the number of clicks, impression, average time spent, search metrics used.
- AWS S3 bucket was used for data import and export.

- Deployed and executed the project in Amazon Elastic Map Reduce (EMR) environment using Scala, SBT, Scala test, HDFS and Map Reduce framework.

Cognizant Technology Solutions, India - Programmer Analyst **Jan 2016 – Dec 2016**

- Worked with the technical team to provide a common integration framework for integrating all the systems in the D&B Enterprise Application.
- Professional experience in SOA, Enterprise application integration (EAI), application software architecture, customer interaction, requirements gathering, analysis, design, and development in Oracle Service Bus platform.
- Proficiently worked with Web Services and REST API to integrate Web-based applications using XML, SOAP and JSON.
- Experience in connecting confluent Kafka topics for transaction logging purpose.
- Interactions with databases like Oracle, SQL through database nodes and structured query language.
- Compiled, organized, analyzed, and presented D&B data which are supplied as risk management, sales and marketing products.
- Coordinated with global teams to achieve overall quality and timeliness of the project deliverables.
- Experience in methodologies such as SDLC, waterfall, Iterative and Agile/Scrum

SKILL SUMMARY

- **Languages** **:** Python, Java, Scala, Bash/Shell, XQuery, SQL, Html, Xml, Json, Yaml, C, C++.
- **Operating Systems** **:** Linux, Windows 10/7, Mac OS.
- **Databases:** **:** Oracle, MySQL.
- **DevOps and SCM** **:** Jenkins, Electric Commander, Bitbucket, Git, GitHub.
- **Container Platform** **:** Docker, Kubernetes, Rancher
- **ML Technologies/Libraries:** Statistical Analysis, Linear/Logistic Regression, Classification, Clustering, Deep Learning, Jupiter, Pandas, Seaborn, Matplotlib, Scikit-learn, Scikit-image, BeautifulSoup, NumPy, NLTK, Tensorflow, Moogsoft, AWS SageMaker.
- **Cloud/Big Data & Others:** Hadoop, HDFS, Spark, SBT, HortonWorks, AWS EMR, CloudSim, Splunk, Grafana, Maven, Putty, JIRA, SOAP UI, Eclipse, IntelliJ, Visual Studio.

ACADEMIA

University of Illinois, Chicago - Master of Science, Computer Science (GPA: 3.56/4)
Coursework: Distributed Objects for Cloud Computing, Data Science, Machine Learning, Artificial Intelligence, Computer Algorithms, Data and Text Mining, Information Retrieval, Neural Network, Advanced DBMS.
Anna University, India - Bachelor of Engineering, Computer Science (CGPA: 8.56/10)
Coursework: Data Structures, Computer Programming, Database Management System, Object Oriented Programming, Computer Architecture, Design and Analysis of Algorithms, Statistics and Probability, Software Engineering, Operating Systems.